

- More generally, there is an **infinite population** perspective that motivates probabilistic modeling for varying coefficients
- Consider the students-schools recurring example, where, e.g., different teaching methods are applied to different schools and we wish to evaluate their effect on students' final exams
- The treatments effects are fixed characteristics and are treated as fixed effects

- On the other hand, there is not specific interest on the students used in the study but rather on the broader population of students
- Hence, student varying intercepts are treated as random effects
- See the example discussed in Section 2.1 in ¹

- From a **regularization** perspective, it is desirable to constraint coefficients for which there is little information in the data
 - For large categorical covariates, for example those that correspond to multi-way interactions, there will be levels that are very sparsely, if at all, observed in the training dataset
- Treating varying coefficients as random and modeled using Gaussian distributions has direct links with ridge regression (but also more general) penalties used in a penalized likelihood framework

- From a **probabilistic inference** perspective, it is natural to model as random variables parameters whose number increases with the observed data especially when the amount of information per such parameter remains bounded, or increases slowly, as total sample increases
- Such quantities are typically called **latent variables**
- The student varying effects is the closest to this situation in the examples we have discussed so far

- But deep interactions, and other ways to build flexible models in this framework, are in the same vein
- The analyst will include deeper interactions with larger sample sizes, as there will be more opportunity to reduce model mis-specification and to do sub-group analysis

- Finally, ¹ offers an **analysis of variance (ANOVA)** perspective
 - “For the purpose of this paper, we identify ANOVA with the structuring of parameters into batches—that is, with variance components models”
 - We will also use the term **variance component** to refer to each group of coefficients, e.g., the age main effects, or the age-state interaction terms
- E.g., see Section 2.2.2 in the article for doing ANOVA with a random coefficient model (in a situation of confounding similar to that of student-schools discussed earlier)

- In summary, we list below some main attractions to random coefficient modeling, providing some explanations for points not covered in the discussion above
- A statistical framework for dealing with sparsely observed levels, especially since uncertainty about those can be integrated when making inferences about fixed effects
 - This allows us to incorporate deep interactions and other categorical covariates with large number of levels
- Implicit modeling of the dependence across repeated measurements (e.g., repeated test results by the same student)

- A probabilistic alternative to identifiability constraints
 - Modeling varying coefficients as random implies a soft constraint on their values
 - For example, the random intercept model $\alpha_j \sim \mathcal{N}(0, \sigma^2)$ for $j = 1, \dots, G$, implies a distribution on $\sum_j \alpha_j / G$ concentrated around 0, with higher concentration for larger G . Therefore, we get a soft sum-to-0 constraint
 - For this reason it is fairly straightforward to learn a model with nested covariates and interactions without having to impose - at least *a priori* - identifiability constraints

- Qualitative and quantitative analysis of sources of variation
 - A main aspect of the grouping of coefficients is that they collectively define a source of variation, e.g., propensity to go and vote due to age
 - Random coefficient modeling provides convenient tools to measure the importance of such source to the model
 - One obvious choice is the estimated variance hyperparameters associated to the group, but see Section 3 in ¹ for more discussion on this

An illustration: fixed vs random effects

- We return to the simple model for Application 1
- Here, we fit it instead with random coefficients using `lme4`¹, a popular library for mixed models
 - We will discuss later the computational technology used in this, it is based on **Laplace approximations** and **sparse linear algebra (SLA)** calculations

```
library(lme4)

glmer(formula = vote ~ scale(z_incstt) +
      z_trnprv + (z_trnprv |
age), data=gg_dat, family=binomial)
```

```
Generalized linear mixed model fit by maximum likelihood  
(Laplace Approximation) [glmerMod]  
Family: binomial ( logit )  
Formula: vote ~ scale(z_incstt) + z_trnprv + (z_trnprv | age)  
Data: gg_dat
```

AIC	BIC	logLik	deviance	df.resid
95307.7	95363.4	-47647.8	95295.7	79142

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.3139	-1.0786	0.5627	0.6754	1.1371

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
age	(Intercept)	0.182753	0.42750	
	z_trnprv	0.002099	0.04582	1.00

Number of obs: 79148, groups: age, 4

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.762650	0.207396	3.677	0.000236	***
scale(z_incstt)	-0.006319	0.008593	-0.735	0.462080	
z_trnprv	0.442686	0.028821	15.360	< 2e-16	***

```
> ranef(mm_model00)
$age
  (Intercept)      z_trnprv
1 -0.67437733 -0.072277046
2 -0.06096816 -0.006534322
3  0.34427016  0.036897489
4  0.39089992  0.041895079
```

```
glm(formula = vote ~ scale(z_incstt) +
as.factor(age)*z_trnprv, family = binomial, data
= gg_dat)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.087650	0.016118	5.438	5.39e-08	***
scale(z_incstt)	-0.006383	0.008594	-0.743	0.4576	
as.factor(age)2	0.613787	0.021467	28.592	< 2e-16	***
as.factor(age)3	1.019630	0.021293	47.886	< 2e-16	***
as.factor(age)4	1.066750	0.026286	40.583	< 2e-16	***
z_trnprv	0.361567	0.035099	10.301	< 2e-16	***
as.factor(age)2:z_trnprv	0.091889	0.045941	2.000	0.0455	*
as.factor(age)3:z_trnprv	0.110529	0.045562	2.426	0.0153	*
as.factor(age)4:z_trnprv	0.117830	0.056919	2.070	0.0384	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 98696 on 79147 degrees of freedom
Residual deviance: 95265 on 79139 degrees of freedom
AIC: 95283

Number of Fisher Scoring iterations: 4

Some observations

- The random coefficient estimates are posterior expectations of the corresponding coefficient with hyperparameters fixed to optimized values
- `lme4` allows for bivariate random coefficients and in the example we have random intercepts and slopes, and by default they are modeled with a prior covariance
 - Here, but apparently it is common, the correlation is estimated to be 1

- `lme4` implements a (hard) sum-to-0 constraint on the posterior means of the random effects
 - This makes the intercepts not directly comparable to those from `glm` that uses pivots
 - For example, the global intercept in `glm` is the intercept for pivot category (age group 18-29), and it is 0.09, whereas to obtain the analogous estimate from `lme4` we take $0.76 - 0.67 = 0.09$
 - Similarly, in `glm` the difference in intercepts between 2nd and 1st age groups is 0.61, and from `lme4` we have $-0.06 + 0.67 = 0.61$

- Generally, the estimates between the two models are almost identical, which is not surprising since there is a lot of data per effect in this case and the regularization provided by the distribution imposed on the varying coefficients is very mild
- The estimated variances of the two groups also match what the fixed effect analysis revealed, that the `z_turnprv` and `age` interaction does not explain any significant variation

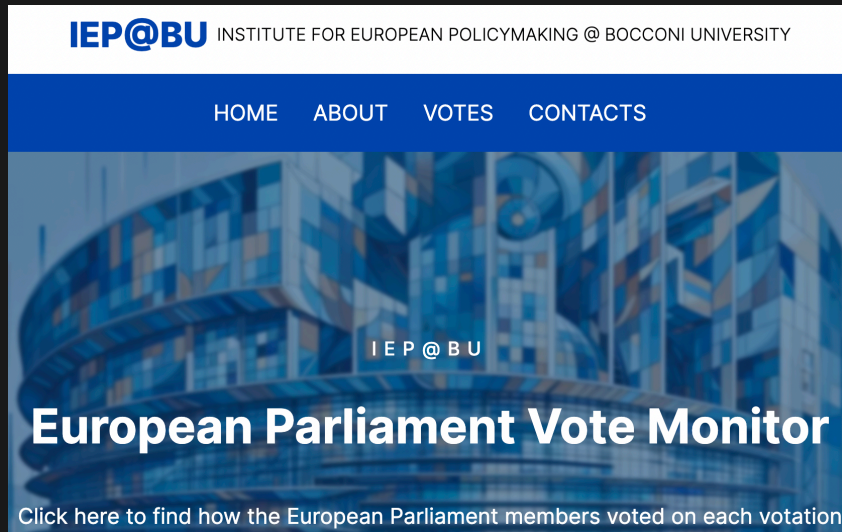
Random factors

Setting

- We move now to understand the second term in the bilinear predictor that involves **random factors**
- This is a fundamental model structure that it is used in many areas of science, including:
 - psychology, political science, economics, demography, and it is pervasive in statistics and machine learning

- For this reason, the concept can be motivated in various different ways, and there are interesting results, model structures and methods scattered around application areas
- From our perspective we will see this directly linked with interactions between categorical covariates and random coefficient models, as developed earlier

Motivating application: ideology in EP



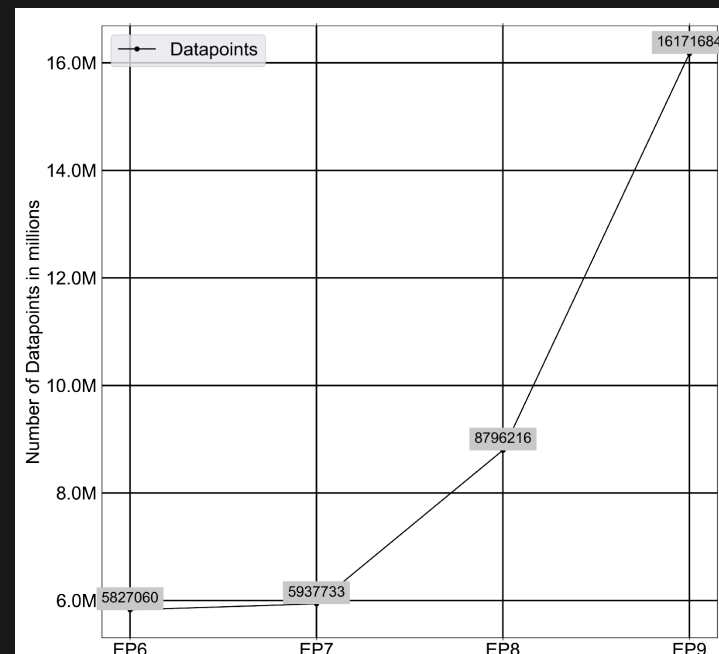
- <https://epvm.iep.unibocconi.eu/>
- Original results from our current research (both in stats/ML and political science)

The data

- EPVM as of 2024, a successor and streamlined version of VoteWatch (Hix & Noury)
- Roll-call and covariates on Members of European Parliament (MEPs) and bills
- Fluidity on European Parliamentary Group (EPG) affiliation (internally and externally)

Data from European Parliament 6 formatted for roll-call scaling

FullName	EPG	1	2	3	...	6200
Adamos ADAMOU	GUE-NGL	1	1	1	...	3
Filip ADWENT	IND/DEM	3	3	3	...	0
Vittorio AGNOLETTO	GUE-NGL	3	3	1	...	3
Gabriele ALBERTINI	EPP-ED	3	1	1	...	3
Jim ALLISTER	NI	3	1	2	...	3
...	...	...	...	...	...	...
Liene LIEPIŅA	EPP-ED	0	0	0	...	3



- Main categorical covariates:
 - MEP-related: MEPD id (**MEP**), EPG (**EPG**), but also country, sex, etc
 - Vote-related: vote id (**vt**), but also content-related, rapporteur, etc
- The matrix data structure “members-votes” is known as **roll call** data
- Notice, compared to previous examples, the fundamental difference that here each crossing of the two categorical covariates is observed at most once

One-mode visualizations

- Create a **co-voting network** of MEPs
 - E.g. two MEPs are connected if they have voted the same way “enough” times; we adapt existing methodology for co-sponsorship networks¹
- We show 2-d maps of **latent** positions of MEPs according to a **force-directed placement algorithm**²
 - Any aesthetics (e.g., EPG affiliation) are added ex-post